

# TESTING

### 6.1 Project Testing

- **Core Functionality:** The application detects Major Depressive Disorder (MDD) in patients using EEG signal features and machine learning models. It provides a Stream lit-based user interface where users can upload EEG feature data or enter specific signal metrics. The system returns a prediction indicating the likelihood of MDD.
- **Model Integration and Input Handling:** Pre-trained machine learning models (e.g., SVM, Random Forest, or Neural Networks) are loaded using the pickle module. Inputs such as EEG frequency bands (alpha, beta, theta, delta) and derived features (e.g., entropy, asymmetry indices) are accepted via structured forms. Input validation is limited and requires improvement to handle missing, incorrectly formatted, or out-of-range values
- **Testing Strategies:**
  - **Unit testing:** Validates the loading of models, correct parsing and processing of EEG features, and consistency of predictions
  - **Integration Testing:** Ensures end-to-end flow from EEG data input to prediction output, including interaction between frontend (stream lit) and backend (ML model).
- **Performance and Error Handling:**
  - **Performance testing:** Measures prediction latency and verifies the system can handle multiple prediction requests concurrently without degradation
  - **Error Handling:** Simulates missing or corrupted model files, incorrect EEG input formats, and system exceptions to ensure meaningful feedback is shown and the app doesn't crash
- **User Experience:** Evaluates the clarity and relevance of EEG input labels, the helpfulness of tooltips or help texts, and ease of navigation in the interface. Testing highlights areas needing improvement in input validation, error messages, and overall accessibility for users without a technical background
- **Performance and Scalability:** If using cloud resources or external EEG feature processing APIs, assess their response under stress. Ensure the feature extraction module scales well with batch EEG inputs or streaming data
- **User Experience:** Test how intuitive the EEG data upload or manual entry process is Check system feedback for incorrect or low-quality EEG input data.

## 6.2 Dataset of healthys.csv for Testing

These are the attributes used to predict the health of a patient.

| name   | type | units | low_cutoff |       | high_cutoff          | description | sampling_frequency |     |
|--------|------|-------|------------|-------|----------------------|-------------|--------------------|-----|
| Fp1    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Fp2    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| F7     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| F3     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Fz     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| F4     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| F8     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| FC3    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| FCz    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| FC4    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| T7     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| C3     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Cz     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| C4     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| T8     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| CP3    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| CPz    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| CP4    | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| P7     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| P3     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Pz     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| P4     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| P8     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| O1     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Oz     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| O2     | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| VPVA   | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| VNVB   | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| HPHL   | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| HNHR   | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Erbs   | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| OrbOcc | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |
| Macc   | EEG  | V     | 0.0        | 250.0 | ElectroEncephaloGram | 500.0       | good               | n/a |

Fig no. 6.1 Health dataset